# Exp:-7

# Lexical Analyzer

```
%%bash
apt-get -qq update
apt-get -qq install -y flex gcc

cat > lexer.l <<'EOF'
%{
#include <stdio.h>
%}

%option noyywrap

%%
"int"|"float"|"char"|"double"|"if"|"else"|"while"|"for"|"return" {
   printf("%s -> KEYWORD\n", yytext);
}

[0-9]+ {
   printf("%s -> NUMBER\n", yytext);
}

[a-zA-Z_][a-zA-Z0-9_]* {
   printf("%s -> IDENTIFIER\n", yytext);
}
```

```
"=="|"!="|"<="|">="|"&&"|"||"|"++"|"--" {
    printf("%s -> OPERATOR\n", yytext);
}

[+\-*/%=<>] {
    printf("%s -> OPERATOR\n", yytext);
}

[\{\}\(\)\[\];,] {
    printf("%s -> SPECIAL SYMBOL\n", yytext);
}

[ \t\n\r]+ {
    /* Ignore whitespace */
}

. {
    printf("%s -> UNKNOWN\n", yytext);
}
%%

int main(void) {
    printf("Lexical Analyzer Output:\n");
    yylex();
    return 0;
}
EOF

flex lexer.l
gcc lex.yy.c -o lexer
```

```
printf 'int a = 10;\nif (a > 5)\n    a = a + 1;\nwhile (i <= 10) { i++; }\nint x = 5 @ 3;\n' | ./lexer
```